OKEKE SAMSON MICHAEL

Machine Learning Engineer and Data Scientist
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PROFESSIONAL PROFILE

Machine Learning Engineer and Data Scientist with 3+ years of experience across predictive analytics, forecasting, fraud detection, customer intelligence, and applied AI. Combines end-to-end model development and deployment skills with technical education leadership at Erikodx Technologies and its training arm, ErikodxIT. Develops practical Python, data science, and machine learning curricula; mentors learners and instructors; coordinates project-based cohorts; and translates technical solutions into business outcomes. Open to AI/ML engineering, data science, technical leadership, and related roles. Currently completing an MSc in Big Data Analytics at the University of Derby.

TECHNICAL EXPERTISE

Machine Learning: Scikit-Learn, XGBoost, LightGBM, CatBoost, TensorFlow, Keras, PyTorch, Statsmodels, time series forecasting (ARIMA, Prophet, LSTM), classification, regression, clustering, NLP, recommendation systems

Model Development: Feature engineering, hyperparameter tuning (GridSearchCV, Optuna), cross-validation, model evaluation (ROC-AUC, precision-recall, F1), explainability (SHAP, LIME), A/B testing

MLOps and Data Engineering: Docker, FastAPI, Flask, CI/CD, model monitoring, MLflow, Airflow, Apache Spark (PySpark), Databricks, ETL pipelines, AWS SageMaker, Azure ML

Data and Analytics: Python (Pandas, NumPy, SciPy), SQL (PostgreSQL, MySQL), R, statistical analysis, hypothesis testing, stakeholder reporting, technical documentation

Visualization: Power BI, Tableau, Plotly, Matplotlib, Seaborn, Streamlit, Excel

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Technical Training & Programme Delivery
Nov 2025 - Present

Erikodx Technologies / ErikodxIT | Port Harcourt, Nigeria

- Develop machine learning workflows for data preprocessing, feature engineering, customer segmentation, forecasting, and performance reporting.
- Design and deliver hands-on instruction and learning materials in Python, data analytics, data science, machine learning, and AI; structure weekly projects, assessments, and capstone activities.
- Support learner onboarding, cohort administration, instructor coordination, project reviews, and completion processes for ErikodxIT training programmes.

- Mentor junior practitioners and students on reproducible experimentation, model evaluation, deployment, code quality, and translating ML concepts into usable applications.
- Founded and run ErikodxIT, a cohort-based technology training platform offering Data Analytics, AI, Power BI, Excel, SQL, and Python.
- Built and deployed the platform: a Next.js website with cohort enrollment pages, and a Moodle LMS (learn.erikodxit.com) with live classes through BigBlueButton.
- Developing a Databricks and PostgreSQL data pipeline for enrollment and sales data, with cohort analytics (financials, enrollment funnel, student demographics) delivered through Power BI and Excel.
- Contribute to expansion planning for ErikodxIT's technology-training portfolio, including new programmes, curriculum structure, and external training partnerships.

Data Scientist | Machine Learning Engineer

Jan 2024 - Oct 2025

Loctech Nigeria | Port Harcourt, Nigeria

- Developed forecasting models across 150+ product categories; reported 92% forecast accuracy and reductions in stockouts (33%) and overstock (28%).
- Built ensemble fraud detection models using transaction and behavioral features; reported 97% accuracy and 95% precision.
- Developed churn models identifying at-risk customers with 85% accuracy, supporting retention initiatives associated with a reported 21% improvement.
- Built recommendation, segmentation, and dynamic-pricing models; communicated insights to stakeholders using explainability tools and dashboards.
- Applied feature selection, experimentation, cross-validation, and model evaluation to improve analytical rigor and operational decision-making.

Data Scientist Intern

Aug 2023 - Dec 2023

Pickmeup Technologies NG | Port Harcourt, Nigeria

- Analyzed large customer and ride datasets with Python and SQL; built predictive and clustering models for customer behavior and segmentation.
- Automated preprocessing workflows and developed dashboards for monitoring model performance and business KPIs.
- Reported 87% accuracy on customer behavior prediction and a 40% reduction in data preparation time.

PROJECTS

- **Hospital Readmission Prediction, MSc Dissertation (University of Derby):** Building models to predict heart failure readmission using the MIMIC-IV clinical dataset, with a full dissertation and reproducible Jupyter notebook.

- **Sales Forecasting and Anomaly Detection Pipeline (Azure):** Production-style ML engineering project using ErikodxIT enrollment data, covering Kafka, Spark, MLflow, FastAPI, Airflow, Streamlit, and Kubernetes.
- **ErikodxIT Learning Platform (Next.js and Moodle):** Built the training platform end to end, including cohort enrollment pages, the Moodle LMS at learn.erikodxit.com, and BigBlueButton live sessions.
- **ErikodxIT Custom Chatbot (in development):** Designing a fully self-owned chatbot in PyTorch with no dependence on external AI APIs.
- **Computer Vision Fruit Classification Application:** Built and iterated on a Keras image-classification workflow, including model training, fine-tuning, evaluation, metadata/catalog generation, and application integration.
- **Customer Lifetime Value Prediction:** Developed regression-based customer value scoring with engineered behavioral and transactional features and FastAPI-based inference.
- **Demand Forecasting & Inventory Optimization:** Combined time-series forecasting and inventory planning methods to support reorder and safety-stock decisions.
- **Real-Time Fraud Detection:** Designed transaction classification and anomaly-focused feature engineering with model monitoring and explainability.
- **Customer Support Text Classification:** Applied transformer-based text classification concepts to automated support-ticket categorization and sentiment analysis.

EDUCATION

MSc Big Data Analytics In progress
University of Derby

B.Sc. Physics and Astronomy Jun 2021
University of Nigeria, Nsukka

CERTIFICATIONS

- **Certified Applied Data Scientist** - WorldQuant University
- **Machine Learning Specialization** - Stanford Online (Coursera)
- **Deep Learning Specialization** - DeepLearning.AI (Coursera)
- **IBM Data Analyst Professional Certificate** - IBM (Coursera)

ADDITIONAL LEADERSHIP & COMMUNITY CONTRIBUTIONS

Technical mentorship, applied ML workshops, curriculum and assessment development, instructor support, educational project repositories, and technical communication for early-career data and AI practitioners.